

2920/201
SYSTEMS ANALYSIS AND DESIGN
July 2020
Time: 3 hours



THE KENYA NATIONAL EXAMINATIONS COUNCIL
DIPLOMA IN INFORMATION COMMUNICATION TECHNOLOGY

MODULE II

SYSTEMS ANALYSIS AND DESIGN

3 hours

INSTRUCTIONS TO THE CANDIDATES

*This paper consists of **EIGHT** questions.*

*Answer any **FIVE** of the **EIGHT** questions in the answer booklet provided.*

*Candidates should answer the questions in **English**.*

This paper consists of 4 printed pages.

Candidates should check the question paper to ascertain that all the pages are printed as indicated and that no questions are missing.

1. ✓ (a) Define each of the following terms as used in *systems theory*:
- (i) functional decomposition;
 - (ii) functional cohesion;
 - (iii) feed-forward. (6 marks)
- (b) Outline **four elements** of the feasibility study report. (4 marks)
- (c) Joy has been employed as a database administrator for a certain company. Outline **four** duties that she is likely to perform. (4 marks)
- (d) Describe each of the following levels of management in an organization with respect to the type of planning:
- (i) strategic;
 - (ii) operational;
 - (iii) tactical. (6 marks)
2. (a) Explain **three** possible causes of ICT *project failure* during systems implementation. (6 marks)
- (b) Table 1 shows different activities of an information system project. Use it to answer the questions that follow.

Activity	Precedence	Duration (Weeks)
A	-	2
B	-	3
C	A	1
D	B	3
E	C,D	3
F	E	4
G	F	3

Table 1

- (i) Draw the network diagram. (5 marks)
 - (ii) Identify the critical path. (2 marks)
- (c) Outline **three** reasons why the *terms of reference* is an important document during systems development. (3marks)
- (d) Differentiate between *modular design methodology* and *computer aided software engineering* tool as applied in systems design. (4 marks)

3. (a) A systems analyst for a certain organization has been requested to come up with types of *documentations* for a proposed system. Explain the **three** documentations that he could generate. (6 marks)
- (b) Differentiate between each of the following pairs of terms as used in network analysis:
 (i) *earliest start time* and *earliest finish time*;
 (ii) *total float* and *free float*. (8 marks)
- (c) Explain **three** circumstances that would necessitate the use of a *questionnaire* as a fact finding method during feasibility study of a proposed system. (6 marks)
4. (a) With the aid of an example in each case, outline **two** types of *system review procedures* in systems development. (4 marks)
- (b) Explain **two** advantages of *parallel changeover* method as used in systems development. (4 marks)
- (c) System analysts use *flowcharts* during systems design. Explain **three** advantages of using this tool. (6 marks)
- (d) A certain company has opted to train its users before the new system is implemented. Explain **three** benefits of this training to the company. (6 marks)
5. (a) Outline **four** advantages of using *Joint Application Design (JAD)* technique in systems development. (4 marks)
- (b) A systems analyst for a certain company intends to perform systems maintenance. Explain **three** types of maintenance that he could perform. (6 marks)
- (c) Jack Company intends to acquire a *transaction processing system* for its operations. Explain **three** characteristics of this system. (6 marks)
- (d) Swivel Company *outsourced* a *data entry module* for its Information Systems. Explain **three** benefits the company would derive from this move. (4 marks)
6. (a) Describe each of the following system design methods:
 (i) Jackson System Development (JSD);
 (ii) Structured Systems Analysis and Design Methodology (SSADM). (4 marks)
- (b) Differentiate between *authorization* and *authentication* as used in systems security. (4 marks)
- (c) Explain **three** disadvantages of using the *waterfall method* during systems development. (6 marks)
- (d) A systems analyst used the *prototyping method* to design proposed information system. Explain **three** advantages of using this method. (6 marks)

7. (a) Outline **four** factors that a systems analyst should consider when choosing a *file organization* method during systems design. (4 marks)
- (b) Describe each of the following types of feasibility studies in systems development:
- (i) legal;
 - (ii) operational;
 - (iii) technical. (6 marks)
- (c) Define the term *structured walkthrough* as used in systems analysis and design. (2 marks)
- (d) Read the following scenario and use it to answer the question that follows.

Jokey airline intends to automate its booking services. The system is expected to allow a new customer to register on the airline portal by keying their details. The customer details are then stored in the customer file. The customer is then issued with a username and password for accessing the airline's portal, which are first validated through a message sent to the customer's phone. Once validated, the customer details are then stored in validated customer file.

After validation a customer is required to login to the portal to select the flight details which may include the travel date, travel time, travel class, place of origin and destination. The selected information is stored in the reservation file for confirmation by the flight attendant at the airline. Payment is supposed to be online on the airline's portal using the customer details previously provided.

Once the payment is processed, the customer is issued with an electronic receipt that is sent to his/her email address to be produced on arrival at the airline.

Draw a level 1 DFD to represent this scenario. (8 marks)

8. (a) Joytech Company intends to acquire an information system using the *off the shelf method*. Explain **three** benefits of this method of acquisition. (6 marks)
- (b) Distinguish between *user requirement* and *functional requirement* as used in systems analysis. (4 marks)
- (c) Explain **three** benefits of using *entity relationship diagrams* in systems design. (6 marks)
- (d) Differentiate between *formative* and *summative* evaluation as used in systems development. (4 marks)

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